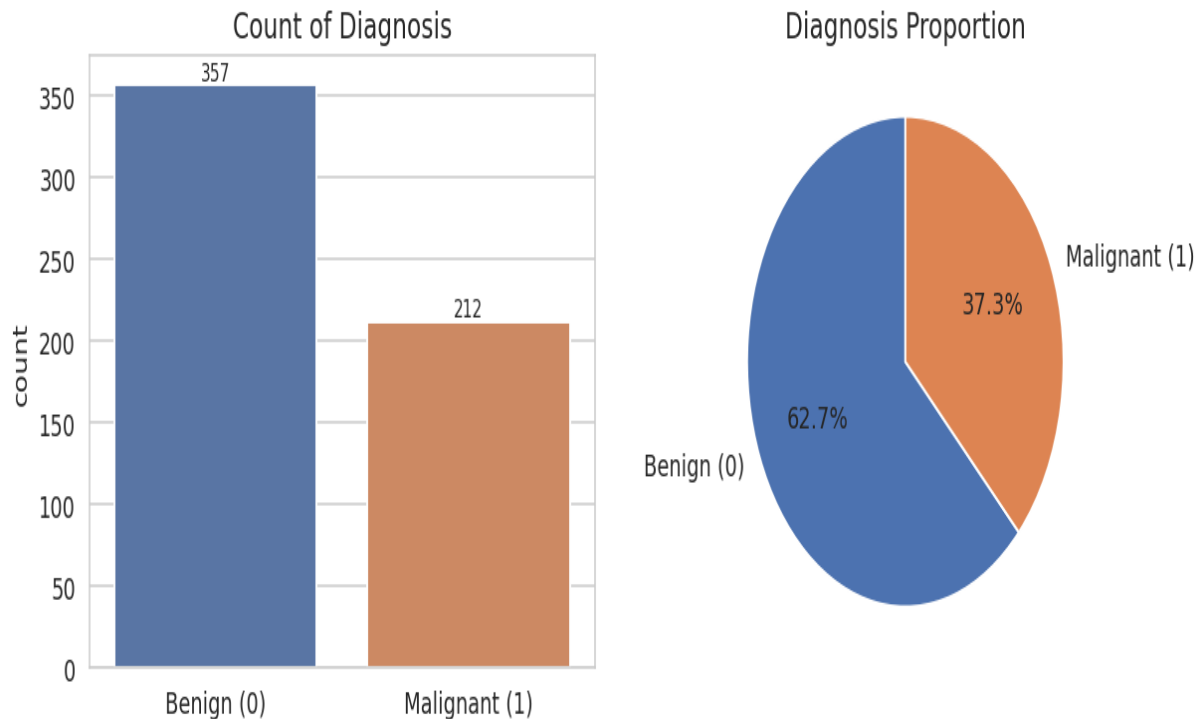


Breast Cancer Wisconsin Diagnosis

EDA Visual Summary — for Model Training Handoff

This is a quick-reference summary of the visual EDA (Part 2) performed on the cleaned dataset (569 rows, 30 numeric features, no missing values, ID dropped, Diagnosis encoded M=1 / B=0). Full interactive analysis is available in the companion Jupyter notebook.

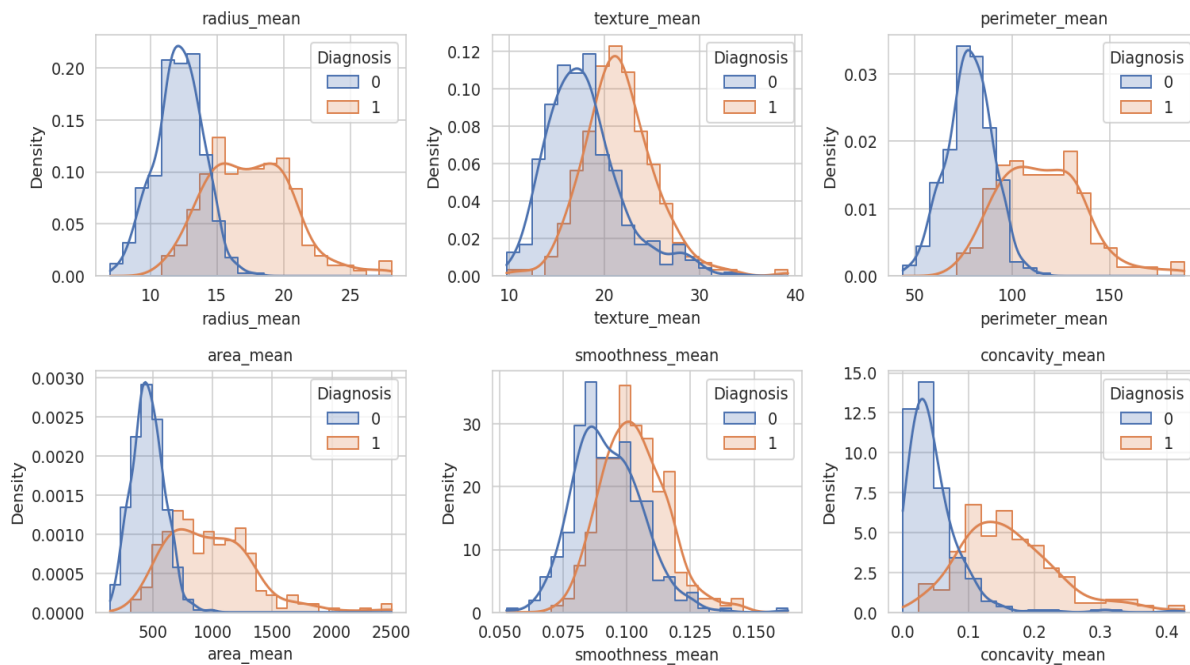
1. Class Distribution



357 Benign vs 212 Malignant (62.7% / 37.3%). Moderate class imbalance — use a stratified train/test split and check precision/recall, not just accuracy.

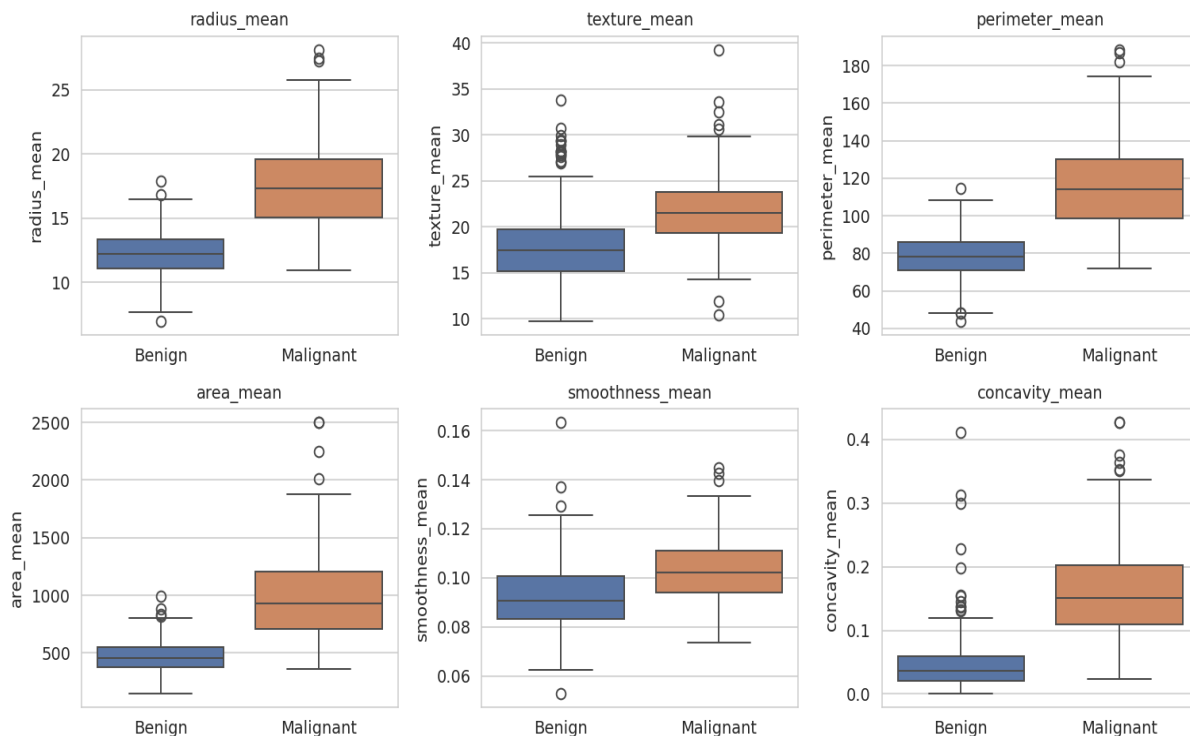
2. Feature Distributions (Histograms)

Feature Distributions: Benign vs Malignant



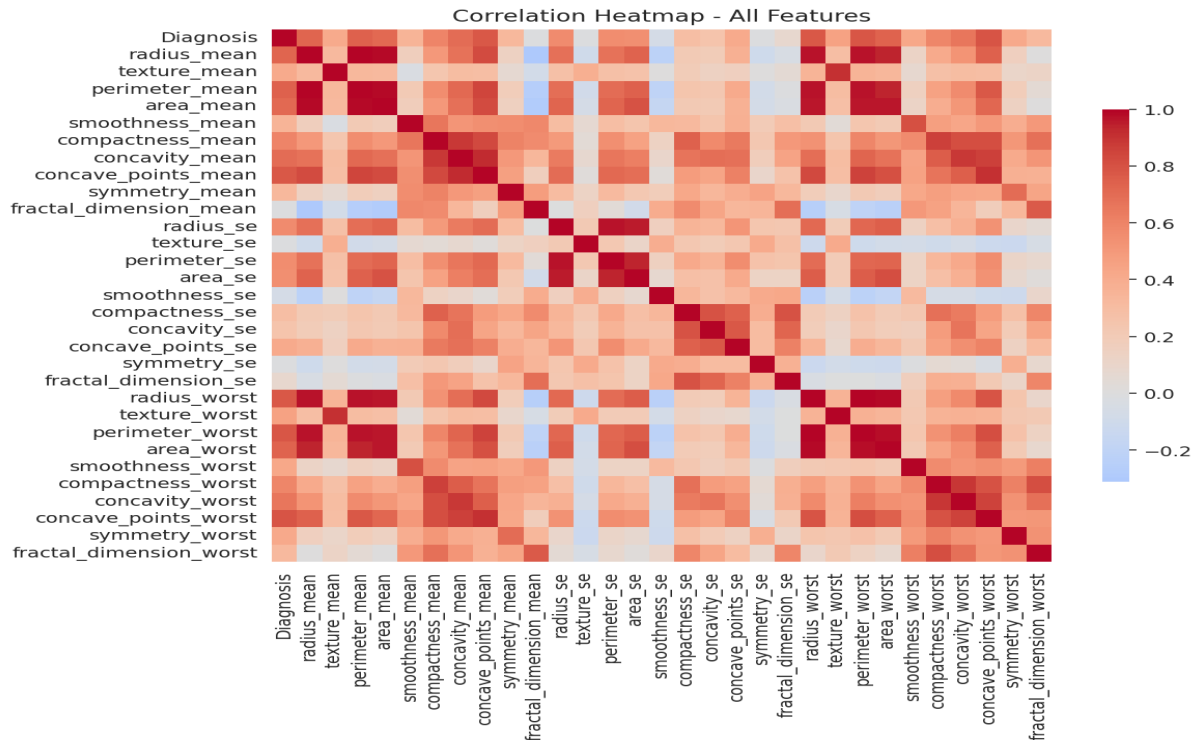
Malignant cases skew higher on radius, perimeter, area, and concavity, with visible separation from Benign cases on several features.

3. Boxplots by Diagnosis

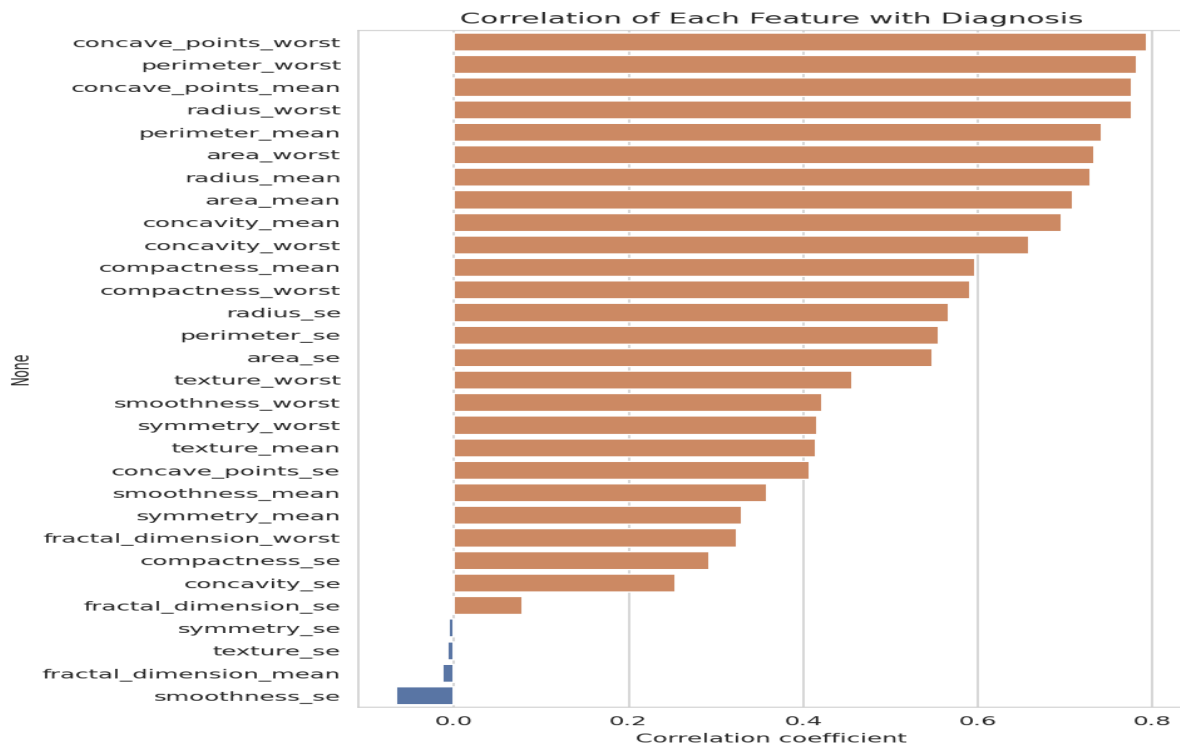


radius_mean, perimeter_mean, area_mean, and concavity_mean show the clearest separation between classes and are good candidate predictors.

4. Correlation Heatmap (All Features)



5. Feature Correlation with Diagnosis

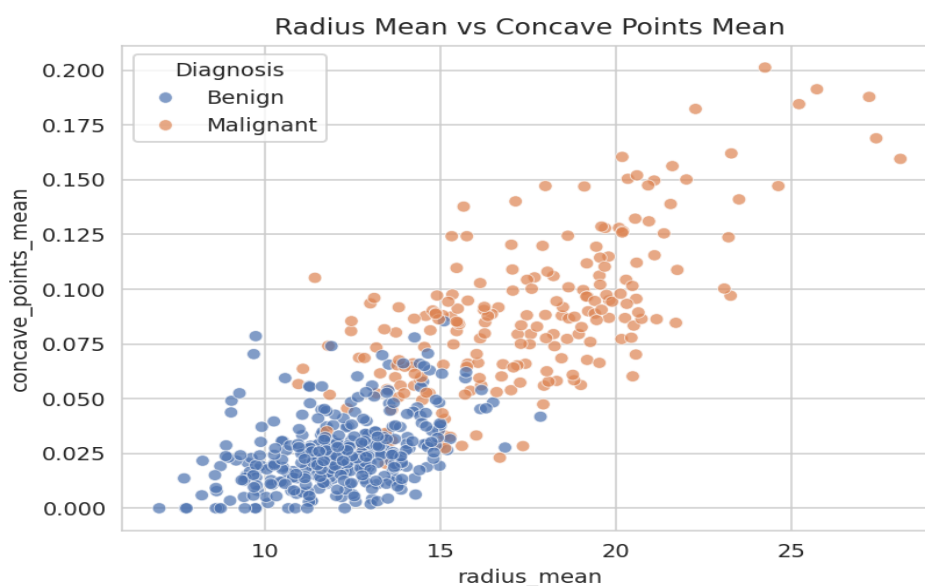


Ranks every feature by correlation strength with the target label.

Top 5 Features Most Correlated with Diagnosis

Feature	Correlation
concave_points_worst	0.794
perimeter_worst	0.783
concave_points_mean	0.777
radius_worst	0.776
perimeter_mean	0.743

6. Radius vs Concave Points (Top 2 Predictors)



Fairly clear visual separation between classes on just these two features — a good early sign for a linear model like Logistic Regression.

Key Takeaways for Model Training

- Moderate class imbalance (~63% Benign / 37% Malignant) — stratify splits, watch precision/recall.
- No missing values — no imputation needed.
- radius, perimeter, area, concavity, and concave points (mean & worst) are the strongest predictors.
- High inter-feature correlation exists (mean/se/worst of the same measurement) — consider this for multicollinearity when interpreting coefficients.